

Skills and employment transitions in Brazil¹

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Abstract:

This paper introduces two methods to derive internationally comparable skills and occupational distance based on machine learning and natural language processing techniques. We apply these measures to analyse employment transitions and workers' wage distribution in Brazil using the universe of formal labour contracts covering the period from 2003 to 2018. Our findings show that workers who use non-routine cognitive skills intensively are better off in terms of employment, wages, and occupational switching. Overall, we observe signs of a routine-biased technological change and employment polarisation, however, this trend has only emerged since the Brazilian economic crisis of 2014.

Keywords: skills; NLP; tasks; unemployment; polarization

Resumo:

Este artigo apresenta dois métodos para derivar habilidades internacionalmente comparáveis e distância ocupacional com base em aprendizado de máquina e técnicas de processamento de linguagem natural. Aplicamos essas medidas para analisar as transições de emprego e a distribuição salarial dos trabalhadores no Brasil usando o universo de contratos formais de trabalho cobrindo o período de 2003 a 2018. Nossos resultados mostram que trabalhadores que usam habilidades cognitivas não rotineiras intensivamente estão em melhor situação em termos de emprego, salários e mudança ocupacional. No geral, observamos sinais de mudança tecnológica tendenciosa à rotina e polarização do emprego, mas essa tendência só surgiu a partir da crise econômica brasileira de 2014.

Palavras-chave: habilidades; PNL; tarefas; desemprego; polarização

JEL codes: J23; J24; J31; E24

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Introduction

Labour markets in developed economies are changing profoundly due to technological progress. Machines and computer capital displace routine occupations in the middle of the wage distribution, and thus employment in Europe and the US is polarising due to the routine-biased technological change (RBTC) (Autor, Levy and Murnane, 2003; Acemoglu and Autor, 2011; Goos et al., 2014; Carrillo-Tudela et al., 2016; Cortes et al., 2020). Hence, fears of mass unemployment due to automation and the advancement of artificial intelligence are recurrent (Acemoglu and Restrepo, 2017; Adamczyk et al., 2021). The phenomenon of jobless recoveries aggravates these trends. The current consensus is that those engaged in routine tasks once again tend to be substituted when firms adjust their technology after a crisis (Jaimovich and Siu 2020; Hershbein and Kahn 2018).

Although these phenomena are considered universal, little is known about how technological change, recessions and recoveries affect workers' wages and employment in low- and middle-income countries. Even if the same fundamental economic forces are at work, labour market outcomes may be quite distinct, as countries differ widely in input endowments, technology adoption, position in the global production chain, or workers' education. For example, Graetz and Michaels (2017) find that modern technology is not responsible for jobless recoveries outside the United States. Obviously, the design of public policies that prepare workers for the future and that shield vulnerable groups from being marginalized depends on our correct understanding of the labour market dynamics in each country.

The main reason for the scarcity of skill-based labour market studies in low- and middle-income countries is the lack of adequate skill measures. This paper provides a novel technique to derive skill measures and skill-based distances between occupations. We then apply these measures to study the relationship between skills, employment transitions and wage changes in the formal Brazilian labour market between 2003 and 2018. Our results allow us to judge whether skill-biased technological change and polarization, observed in high-income countries, are also present in Brazil. This country is representative of Latin America and one of the largest among its peers. It provides an interesting case because Brazil experienced both a long period of prosperity with falling inequality and rising wages and an economic slowdown and recession since 2015, deepened by the Covid-19 pandemic. Whether the improved prospects in the Brazilian labour market will continue depends largely on how the 'jobless recovery' and RBTC phenomena apply here (Firpo and Portella, 2019).

The continuous skill measures are derived from the occupational description in the country under consideration; however, the procedure is simple and readily applicable to any other country. As a unique feature, our method replicates rather than relabels existing skill measure definitions. It can thus serve to support studies that are highly comparable across countries. Its main advantages are the modest data requirements and avoidance of cross-country occupation matching or other ad hoc assumptions. Specifically, only two inputs are needed: an international skill standard to be replicated and task descriptions of occupations in the country under consideration. In the present case, we rely on the Brazilian Classification of Occupations (Classificação Brasileira de Ocupações, CBO) and generate the frequently used categories of non-routine cognitive (NRC), routine cognitive (RC), non-routine manual (NRM) and routine manual skills (RM) from Autor, Levy and Murnane (2003). The basic idea of our procedure is to identify the verbs that best characterize the original classification, translate these verbs, identify and weigh the translated verbs in the target country's occupational description using machine learning and natural language processing techniques. Consequently, the method does not assume that occupations in the origin and target country perform the same activities.

In comparison, existing approaches can be classified into three groups: those derived from a combination of the tasks originating from the United States with an occupational crosswalk (Goos, Manning and Salomons 2014; Arellano-Bover 2020; Ehrl and Monasterio 2021); discrete classifications based on an

ad hoc labelling of a few occupations (Bachmann, Cim and Green, 2019); or continuous measures derived from the ad hoc classification of tasks as shares in occupations, in the spirit of Spitz-Oener (2006). Besides the problems introduced by the imperfect occupational matching, any of these previous methods were proposed for developed countries and, thus, are not adapted to skills requirements and tasks characteristics of jobs in low- and middle-income countries. In Brazil, some task-based studies exist, but with different focus and skill measurement methods.² These papers either apply Maciente's (2013) mapping between O*NET and Brazilian Occupational Classification (Classificação Brasileira de Ocupações – CBO) or define discrete skill measures using occupation groups.

Our second methodological contribution is to derive a measure of occupational distance to indicate how different any pair of occupations are in terms of their task contents. This distance variable is applied to evaluate occupational changes in the context of human capital and productivity losses, as previously done by Gathmann and Schönberg (2010). Again, this technique renders country-specific measures and can be applied whenever a national occupational dictionary with task descriptions is available.

For the application of the skill and occupational distance measures, we rely on the administrative registry of all formal employment contracts in Brazil between 2002 and 2019, the RAIS (Relação Anual de Informações Sociais). Taking a 10 per cent random sample from this database leaves us with over 50 million observations of 6.4 million men and women from all sectors. To the best of our knowledge, we are the first to provide nationally representative evidence regarding the relationship between skills and employment transitions in Brazil.

As a first step, we document that cognitive skill scores increase over the wage distribution, and manual skill scores decrease. Additional metrics also suggest that our skill measures are in line with evidence from other countries. Overall, there have been signs of routine-biased technological change and employment polarization, but only since the 2014 Brazilian economic crisis. Middle-skilled occupations in the manufacturing sector show the largest relative employment losses during the recent recession, from 2013 to 2018. Moreover, when occupations are ranked by their mean wage, we observe employment gains at the lower and upper end of the distribution, while occupations in the middle of the wage distribution suffer employment losses.

Previous evidence on job polarisation in Brazil is mixed. Technological advances favoured middle-skilled workers between 1981 and the late 1990s (Herdeiro, Menezes-Filho and Komatsu, 2019). During the period under investigation here, Maloney and Molina (2019) observed some signs of polarisation, whereas Ariza and Bara (2020) report that the employment change curve over the wage distribution is flat. Firpo and Portella (2019) suggest that age-biased technical change or skill obsolescence offset polarisation trends. Only Almeida, Corseuil and Poole (2017) find evidence in favour of polarization, in line with our results. Differences in data, methodological approach, and temporal coverage explain the diverging results. Only one of the cited papers (Almeida et al., 2017) also uses the complete RAIS data, whereas the other relies on survey data such as the PNAD or IPUMS.

Our results also confirm that workers' skills are related to employment transitions. In particular, we observe that workers with intensive NRC skills experience relatively low probabilities of occupational and sectoral switching. Their labour market exit risk is relatively low, and NRC workers tend to find close, higher-paid occupations in the case of an occupational change. Regarding these characteristics, RM and NRM workers are clearly worse off. There is also a tendency for middle-wage workers to use more and more NRC skills. RM skilled workers seem more vulnerable than NRM workers. For example, RM workers show the lowest transition rates to occupations that rely on different, higher-paid skill types. RM workers also tend to move to more distant occupations when workers change occupations or sectors.

Regarding middle-income countries, as far as we know, in-depth studies on occupational transitions are rare. Samaniego and Viegelahn (2021) analyse workers' jobs and occupational switches in Vietnam, and

² For example, Ehrl and Monasterio (2019) and (2021) study the long-term effects of skill concentration in local labour markets. Albuquerque et al. (2019) predict the impact of automation on the Brazilian labour market and Adamczyk, Monasterio, and Fochezatto (2021) on the public sector. Ehrl (2018) analyses the heterogeneous effects of intermediate goods importing on the skill structure within firms.

Puggioni et al. (forthcoming) document the wage penalty for exiting and re-entering the formal labour market in Mexico. Both countries are also characterized by high transition probabilities between formal jobs and either unemployment or informality.

2 Data, variable definitions, and methodology

2.1 Data

The main data source for this research is the Annual Social Information Report (*Relação Anual de Informações Sociais* - RAIS) produced by the Brazilian Ministry of Economics. RAIS is an administrative register of employer-employee data covering the population of formally employed workers in the public and private sectors. It is widely recognized as Brazil's most reliable source of information about the formal labour market (Dix-Carneiro 2014). We draw a 10 per cent random sample from the population. Our analysis includes male and female employees between 20 and 60 years of age with complete information on the selected variables.³

The informal sector represents almost half of the labour force in Brazil. Of course, it would be ideal if our data also included informal workers. Still, studies have shown that the informal sector in Brazil acts as a buffer for the unemployed in the face of adverse labour demand shocks (Dix-Carneiro and Kovak 2019). Like unemployment rates, informality is counter-cyclical, and the transition rates from unemployment to informal jobs are twice as high as those from unemployment to formal jobs (Ulyssea 2020; Dix-Carneiro et al. 2021). In other words, the boundaries between unemployment and the informal sector are quite fluid and more comparable to each other than to formal jobs in the highly regulated formal labour market. Thus, this study deals with the internal dynamics of workers in the formal labour market and its inflows and outflows, irrespective of whether they are unemployed or working in informal activities.

The data are organized in the form of employment spells. That is, the number of observations per worker-year corresponds to the number of non-overlapping formal employment relations. Labour market entry and exit are permitted anytime during the longest possible observation period from 2003 to 2019. The present analysis considers workers' wages, the number of hours worked per week, their occupation, tenure on the job, education level and their employers' size, sector, legal nature and geographical location. Since we are mainly interested in how workers' characteristics relate to employment transitions, the last year is only utilized to calculate change variables (such as occupation change, wage growth and employment distance). The final full sample from 2003 to 2018 contains 50,082,074 observations from 6,358,348 individuals. Besides, left- and right-censoring of employment trajectories require careful treatment. On the one hand, duration analysis (for the probability of leaving the formal labour market) is adequate for this situation. On the other, to analyse occupation changes, we follow the convention in the literature and restrict the sample to workers with at least two consecutive employment spells (e.g., Bachmann, Cim and Green, 2019 or Cortes et al. 2020). In this consecutive employment sample, we end up with 46,923,670 observations from 5,433,062 individuals.

2.2 Occupational distance measure

In order to find a continuous metric for the distance between a pair of occupations, we proceed in two steps. Firstly, we apply Natural Language Processing (NLP) algorithms to find weights that express the relevance of each word in the Activities Matrix,⁴ which describes the tasks inherent to each occupation. Secondly, we choose a distance measure and use the weights from the previous step to derive the occupational distance

³ Because our empirical strategy requires that workers' skills can be related to their employment trajectories, only workers with no more than one job at the same time are considered. Due to the particularities and low transition propensities, members of the armed forces are excluded as well.

⁴ The CBO Activities Matrix contains an average of ten task descriptions for each of the 2,474 different occupations when the most disaggregate 6-digit classification is used. For example, the occupation "truck driver" (code 782510) lists tasks such as "transport container", "load vehicle", "deliver cargo", "handle cargo with safety", "inspect mechanical parts", and so on.

matrix. Previous studies, such as Gathmann and Schönberg (2010), use a distance metric based on the absolute difference between analytical task scores. Instead, we directly compare the weights of words that describe each occupation's tasks.

The Term Frequency–Inverse Document Frequency (TF-IDF) method calculates the relative weights of words describing the tasks. The idea of the TF-IDF is that the derived weights increase proportionally to the frequency of occurrence of a word in a given text extract in relation to the number of texts that contain the word. Following NLP jargon, *corpus* is a collection of documents, document is a text object, and text object is a sequence of words. In this subsection, *corpus* is the set of CBO activities descriptions, and the documents are these descriptions. Given the set of occupations' activities D , a word w in the activity description, $d \in D$, the calculation of the corresponding weight ($weight_{w,d}$) according to the TF-IDF method is given by

$$weight_{w,d} = f_{w,d} \cdot \log \left(\frac{|D|}{f_{w,D}} \right) \quad (1)$$

where $f_{w,d}$ is the relative frequency of w in d , defined as the number of occurrences of each word relative to the total number of words in the activity description d . $|D|$ is the total number of activities descriptions and $f_{w,D}$ is the number of activities in D in which w appears. High values of $weight_{w,d}$ imply that w is an important word in d (first term) but not very common in D (second term) in equation (1). In that sense, high TF-IDF weights signal that the word w serves well to discriminate between the task sets of occupations.

In order to deal with grammatical variations of the same word, a stemming procedure is carried out before applying the TF-IDF method. Therefore, inflexions of the same term are converted to a single common root (e.g., “analyse”, “analyse”, “analysed”, or “analysing” are the same). In addition, stop-word removal discards phrase-connecting words, such as conjunctions, numbers and special characters with no semantic value for the analyses.

The resulting vector, $W_d = (weight_{1,d} \dots weight_{W,d})$, collects the TF-IDF weights of all words from 1 to W , as W being the number of different words that appear in the Activities Matrix. The matrix $W = (W'_1 \dots W'_D)$ ends up with 2,322 words describing 2,641 occupations. This matrix is 99 per cent sparse since most terms are not shared between occupations.

We apply the cosine similarity measure to derive distances between occupations as vectors in high dimensional sparse matrices. The cosine similarity computes the angle between two vectors, that is the weights of words that compose the activities of two given occupations. Because the cosine measure is insensitive to the absolute length of each document, it works well for sparse matrices.⁵ Based on Aggarwal (2015), the cosine similarity for any pair of occupations i and j is defined as

$$Cos_{ij} = \frac{\sum_{w=1}^W weight_{w,i} \cdot weight_{w,j}}{\sqrt{\sum_{w=1}^W weight_{w,i}^2} \cdot \sqrt{\sum_{w=1}^W weight_{w,j}^2}} \quad (2)$$

Therefore, we have a symmetric matrix reflecting the bilateral similarity between occupations. Cos_{ij} is a continuous measure in the interval $[0,1]$, where occupations with value 1 are identical to each other in terms of tasks, and occupations with 0 do not share words describing their tasks. Finally, to have a direct measure of the distance between occupations, we use $OccDist_{ij} \equiv \log(1/Cos_{ij})$ in the following analysis.

2.3 Skill measures

We propose a novel methodology to generate skill measures with three advantages over previously used ones. The first advantage is that we do not need ad hoc lists of words related to each type of task. Second, the method does not assume that the occupations perform the same activities across countries. Third, our

⁵ The Euclidian distance, the most common distance metric in Machine Learning, is not suitable for a high dimensional sparse matrix (Aggarwal, Hinneburg and Keim 2001).

method can measure the skills even of occupations not listed in the origin skill’s database, such as O*NET. A large number of Brazilians work in occupations that cannot be found in O*NET, such as elevator operators in commercial buildings (CBO 5141-05 - Ascensorista) or bus ticket/money collectors (CBO 5112-15 - Cobrador).

We chose to follow the skills categories introduced by Autor, Levy and Murnane (2003), which were then adapted to the O*NET data by Acemoglu and Autor (2011). This will be the only relevant ad hoc decision in the following procedure. In order to replicate the United States’ data-based skills, we use the O*NET as an initial stepping stone. The remainder of the procedure relies exclusively on the Brazilian CBO data. The derivation of skill measures is carried out in five steps:

1. Build a dictionary of verbs from the O*NET that are associated with occupations with the highest skills in each of the groups (NRC, RC, NRM, and NRM);
2. Translate these verbs to Portuguese through a deep learning translation service;
3. Derive weights of these Portuguese verbs employing an NLP technique;
4. Apply these weights by skill to the text tasks descriptions of the CBO occupations;
5. Normalise skill measures.

Our choice of skill measures is the four categories from Acemoglu and Autor (2011): NRC, RC, NRM and RM.⁶ Having calculated the original task measures of Acemoglu and Autor (2011), we rank occupations by their skill score and select the top 10 per cent of occupations. Repeating this procedure for all four skill categories defines the selected occupations’ elements, whose words are translated into Portuguese, resulting in around 6,100 tasks.

The conversion of the task file from English to Brazilian Portuguese is done with *DeepL Pro*, a neural machine-learning translation service.⁷ The tasks’ descriptions contain information about contexts that would be undesirable for our purposes. For example: “clean a car” and “design a car” contain verbs associated with different skills, but the noun “car” would have the same weight as the verbs and would harm the classification. Therefore, we prefer to extract only the verbs from the task descriptions.

In this subsection, we use two corpora: verbs associated with skills in O*NET and the verbs in the activities descriptions by occupation in CBO. From the 96,900 words in the translated task descriptions of O*NET, 996 verbs have correspondence in the CBO task descriptions. The matrix V contains the weights (as defined below) for the 996 verbs (columns) in the four skill categories $i \in \{NRC, RC, NRM, RM\}$ (rows). Our final skill scores for each occupation are the result of the following matrix product:

$$Skill = V * Occup \quad (3)$$

where the matrix $Occup$ is defined by the raw count of the 996 verbs (rows) in the 2,478 CBO occupations (columns).⁸

The usual way to identify the weights of terms in a document for V is through the TF-IDF method (see section 2.2). Here, however, TF-IDF has not turned out to be the best option. The reason is simple: if a verb is present in all four skill categories, the value of the IDF is 0, and obviously, TF-IDF is equal to 0.⁹ For example, if a given verb w_v appears 100 times in total with the following distribution: 90 times in the RM

⁶ Minor adaptations and corrections are made to increase the meaningfulness of skills in the Brazilian context. Appendix A lists the task requirements that define these skill categories.

⁷ DeepL uses a proprietary algorithm with convolutional neural networks. In blind tests, professional human translators considered the DeepL translations better than those by Google, Amazon and Microsoft services (see <https://www.deepl.com/en/quality.html>).

⁸ The number of CBO occupations differ from the previous section because a few CBO occupations contain verbs for which there is no translation from the O*NET verbs. In these cases, the skill scores are missing. To avoid the artificially resulting gaps in workers’ labour market transitions, we replace the missing values with the average skill scores of the corresponding more aggregate 5- or 4-digit occupation classification.

⁹ NLP technical terms may create misunderstandings between the use of the term “document” in this subsection and in the previous one. In measuring the distance between occupations, “documents” referred to activities descriptions. In the skills measurement subsection, it refers to the verbs associated with each of the four skills.

skill category, eight times in NRM, and once both in RC and NRC, the TF-IDF value of the verb w_v will be zero. The same value would be obtained if w_v appeared with equal frequency in the four skills.

To overcome the distortion that TF-IDF would cause, we replace the IDF with the repeat rate (RR) to measure the relevance of the verbs in the CBO occupation descriptions (NLP documents in this subsection).¹⁰ The RR of a verb is the sum of squares of the ratio between the count of verb w_v in each skill and the total count of the same verb. The minimum value of the RR is 0.25 (4×0.25^2) for a verb that is equally distributed among skills, and the maximum value of 1 is obtained if the verb is observed in just one skill.

The main advantage of the TF-RR over TF-IDF is that it is more suitable for asymmetric distributions. If a verb is perfectly distributed across our four skills, then $RR = 0.25$ and $IDF = 0$. In the numeric example above, with a very asymmetric distribution, the RR would equal 0.83 ($0.9^2 + 0.08^2 + 0.01^2 + 0.01^2$) while IDF would still be 0. For the mathematical definition of TF-RR, the reader is referred to the Data Appendix A.A2. Ultimately, the combined TF-RR procedure provides the weights that are used as elements in the matrix V according to the product of TF_{wk} and RR_w . As before, TF is defined as the relative frequency of a verb w_v in the total number of verbs in skill k .

To interpret the descriptive statistics and regressions, skill scores are standardized so that the sum of the four categories equals one in each occupation. The final scores can therefore be interpreted as shares. For example, $NRC=0.25$ means that a quarter of the average workload in this occupation is executed using NRC skills. A characteristic feature of our skill definition is that the scores differences across occupations are relatively small, see Table 1. The reason is that even occupations with the highest NRC scores require some RC activities (such as making phone calls) or RM skills, like typing on a computer.

2.4 Econometric approach

2.4.1 Transition regressions and testable hypotheses

The research question in this paper is: how are skills related to labour market transitions in Brazil? In particular, the following hypotheses will be tested in our regressions and duration analysis: Workers' skill endowments affect the probability of transitioning to a different occupation [H1], sector [H2] or out of the formal labour market [H5]. In case of an occupation change, skill endowments affect the distance between occupations [H3] and the wage differential [H4].

The dependent variables that allow us to assess hypotheses H1 to H5 are defined according to the difference in the employment outcome of worker i in the current period t and the sequential employment spell (which must not necessarily be in year $t + 1$). We use the following variables:

- $\Delta Occ4$ is an indicator variable for whether we observe that a worker has a different 4-digit occupation in the following employment spell.
- $\Delta Occ2$ is defined similarly to the previous definition but based on 2-digit occupations.
- $\Delta Occ4\uparrow$ is an indicator variable for whether we observe that a worker moves to a different 4-digit occupation with a higher average wage. Average wages are defined as the mean log wage for a given year and 4-digit occupation based on the observations in our sample.
- $\Delta Occ4\downarrow$ is an indicator variable for whether we observe that a worker moves to a different 4-digit occupation with a lower average wage.
- $\Delta Sect$ indicates whether a worker will move to a different sector.
- $\log(OccDist)$ is the log of the inverse of the occupational distance defined in equation (2).
- $\Delta Wage$ is the difference between the log wage in the next and the current employment spell.
- $OFLM$ is an indicator variable for whether a worker experiences at least one month out of the formal labour market between the current and the next employment spell.

¹⁰ The repetition rate is identical to the Hirschman-Herfindahl index in Economics or the Simpson (1949) index in Biostatistics.

Our baseline employment transition regressions take the following form

$$y_{it} = \alpha + \beta X_{it} + \sum_k \gamma_k skill_{it}^k + D_s + D_r + D_t + \epsilon_{it} \quad (4)$$

where the vector of control variables X_{it} includes worker i 's number of working hours, level of education, tenure, gender, age and firm size. D_s , D_r , and D_t are dummies for sector, federal state and year, respectively. The coefficients of interest γ_k then indicate how the current skill levels are linked to employment transitions of observably equivalent workers in the near future.

In line with the definitions above, we use NRC, NRM, and RM skills in the regressions, while RC skill is the omitted reference category. If the dependent variable is an indicator, we apply logit regressions; otherwise, equation (4) is estimated with pooled OLS. Standard errors are clustered at the occupation level because the explanatory variables of interest - workers' skills - vary only across occupations. This procedure yields more conservative results than clustering at the worker level.

2.4.2 Duration analysis

Duration analysis estimates the probability of leaving, given that this event has not occurred before, conditional on the set of observable variables (Lancaster 1979). The exit probability at time t upon survival to time t is known as the hazard rate. If the duration time t follows a Weibull distribution, which is the most frequently used in the literature, the hazard rate θ takes the following form

$$\xi Z_{it} = \alpha + \beta X_{it} + \sum_k \gamma_k skill_{it}^k + D_s + D_r + D_t + \epsilon_{it} \quad (5)$$

$$\theta_{it}(t \vee Z_{it}) = pt^{p-1} \cdot \exp(\xi Z_{it}) \quad (6)$$

where Z_{it} is composed of the same control variables defined in equation (4) and p is a parameter that indicates whether the exit probability has an increasing, decreasing or constant form. The parameters of the duration model are obtained by maximum-likelihood estimation, where standard errors are also clustered at the occupation level.

According to the nature of our worker panel, the survival data includes multiple records per subject with continuous-time measurement and the possibility of multiple exits. The exit event is defined as leaving the formal labour market for at least one month. Because we cannot distinguish whether the individual becomes unemployed, self-employed, or gets a job outside of the formal labour market, we consider only exits for individuals that eventually return to the formal labour market. If an individual does not return to the formal labour market, we have no way of knowing the direction of their transition into a lower or higher wage or distant occupation.

3 Results

3.1 Descriptive statistics

3.1.1 Skill scores and workers' characteristics

It is crucial to check whether our novel skill measures produce adequate results. The continuous skill measures' mean values in the most aggregated occupation groups are plausible. There are nine groups, excluding police and military professionals, and taking into account both groups of industrial workers (discrete and continuous processes) in CBO.

► **Table 1. Skill scores by occupations.**

	Manag. (1)	Profes. (2)	Techn. (3)	Admin. (4)	Manuf. 1 (5)	Maint. (6)	Manuf. 2 (7)	Service (8)	Agric. (9)
NRC	0.28 (0.01)	0.29 (0.02)	0.27 (0.02)	0.25 (0.01)	0.21 (0.03)	0.22 (0.02)	0.22 (0.02)	0.24 (0.02)	0.23 (0.03)
RC	0.29 (0.01)	0.28 (0.02)	0.28 (0.02)	0.29 (0.01)	0.25 (0.02)	0.25 (0.01)	0.25 (0.02)	0.27 (0.02)	0.26 (0.02)
NRM	0.24 (0.02)	0.22 (0.01)	0.24 (0.02)	0.24 (0.01)	0.24 (0.03)	0.27 (0.01)	0.27 (0.02)	0.25 (0.01)	0.24 (0.03)
RM	0.19 (0.02)	0.20 (0.02)	0.22 (0.02)	0.22 (0.01)	0.30 (0.05)	0.26 (0.02)	0.27 (0.03)	0.24 (0.03)	0.27 (0.02)

Note: the cells show the mean and standard deviation (in parenthesis) for the skill measures by 1-digit occupations from 2003 to 2018. Calculations are based on the full sample and apply weights for employment duration and hours worked. The 1-digit occupations are ranked according to their mean wage in descending order, namely: directors and managers; science and arts professionals; medium-level technicians; administrative service workers; workers in the production of industrial goods and services (1), workers in repair and maintenance; workers in the production of industrial goods and services (2); service workers and sellers; and agricultural workers.

Table 1 shows that the groups with the highest wages - professionals, managers and technicians - present the highest scores in NRC skills (with the full RAIS sample data from 2003 to 2018). These occupations are responsible for abstract and highly specialized tasks, intensive in hard and soft skills since most work is done by interacting with others. Managers and technicians have the highest scores in RC skills as well. Besides, administrative and service occupations mostly require RC and, to a lesser extent, NRC skills. As expected, the highest scores on NRM skills belong to workers in maintenance, service and manufacturing 2 (continuous processes). Manufacturing 1 (discrete processes) and agricultural workers score highest in RM skills as those workers mainly perform repetitive tasks in controlled environments, with little or no necessity for improvisation.

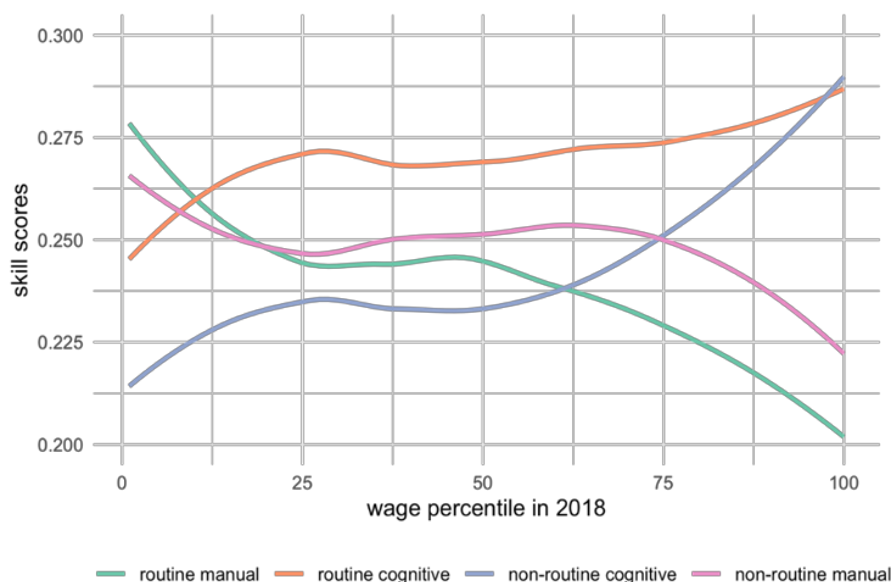
Figure 1 provides another way to compare the plausibility of our skill measures to previous findings from other countries. Following Autor, Katz and Kearney (2008), the figure shows how the skill scores varied across the wage distribution in 2018. Additional information for previous years is provided in the Appendix (see Figure B1). RM and NRM-intensive occupations dominate at the lower end of the wage distribution, exhibiting a negative and decreasing slope over the wage distribution. NRC skills show a mirror-inverted pattern, beginning at a low level and rising at an accelerated slope over the upper half of the wage distribution. RC skills are most frequently used in occupations between the 20th and the 80th percentile. While the RC skill intensity is also increasing in wages, it is almost flat for the most part in the middle of the distribution.

Regarding the evolution over time, Figure B1 in the Appendix indicates that skill intensities of high- and middle-wage workers remained quite constant. The most pronounced change is that low-wage employment now uses fewer NRC skills, whereas NRM skills are becoming more critical. Middle-wage workers also tend to increasingly use NRC skills, in line with the RBTC hypothesis.

It is informative that the skill intensities over the wage distribution are similar to the United States experience. For example, Autor, Katz and Kearney (2008) report that NRC skills are monotonically increasing, NRM skills are markedly decreasing, and RC skills are most frequent in middle-wage occupations. These patterns are as expected and are not related to taking the United States task descriptions in the O*NET as a starting point for our method. The observed patterns in Figure 1 reflect that NRC skills require a large investment in human capital (such as a university degree). Consequently, the return to these skills is highest, and occupations with high NRC requirements are at the top of the wage distribution.

The pattern in Figure 1 and the order of relative wages are in line with other studies of the Brazilian labour market. Controlling for observed and unobserved characteristics in a fixed-effects model, Detoni, Freguglia and Corseuil (2020) find higher returns for NRC-intensive occupations, followed by RC, RM, and NRM skills. This evidence is in line with RBTC; that is, advancing automation substitutes previous routine tasks performed by humans and therefore degrades their wages (Acemoglu and Autor 2011).

► **Figure 1. Skill scores over the wage distribution in 2018.**



To evaluate how the socioeconomic characteristics of workers relate to their skills, we define discrete skill groups following Goos and Manning (2007) and Cortes (2016), among many others. These categories are listed in Data Appendix 1. Table 2 shows the descriptive statistics by comparing the occupational and socioeconomic characteristics of each skill group between 2003 and 2018.¹¹ NRC workers receive higher wages, followed by RC, RM, and NRM workers. This ordering remains stable through 2018, but all occupational groups experienced real gains over our observation period. In particular, NRM and RM occupations gained 33.8 per cent and 29.9 per cent higher real wages than in 2003. This catching up may not be a sign of high demand but rather be influenced by the important adjustments of the minimum wage, which has grown by 76.5 per cent in real terms since 2003. Regarding gender, men dominate the RM occupations in both years, scoring 83 per cent on average. On the other hand, women increased their participation in the three remaining categories and are predominant in NRC, RC, and NRM occupations. The most relevant effect of the increased female labour market participation can be seen in NRM jobs, which had 58 per cent of its positions occupied by men in 2003, decreasing to 47 per cent in 2018. These NRM occupations encompass service and cultural services workers, technicians in transport services and agricultural producers. Table 2. Descriptive statistics by skill groups: 2003–2018.

Table 2. Descriptive statistics by skill groups: 2003–2018.

¹¹ The values in Table 2 refer to the consecutive employment sample that will be used for most of the regression analysis below. Notwithstanding, the full sample yields very similar patterns.

	2003				2018			
	NRC (1)	RC (2)	NRM (3)	RM (4)	NRC (5)	RC (6)	NRM (7)	RM (8)
wage per hour (BRL)	126.34 (164.91)	55.70 (77.16)	31.90 (36.26)	39.71 (40.43)	150.28 (179.78)	65.30 (76.86)	42.68 (36.71)	51.57 (44.09)
male	0.49 (0.50)	0.44 (0.50)	0.58 (0.49)	0.83 (0.37)	0.46 (0.50)	0.40 (0.49)	0.47 (0.50)	0.83 (0.38)
age (years)	36.08 (8.82)	32.52 (8.81)	34.98 (8.97)	33.47 (8.77)	39.76 (9.32)	37.91 (9.89)	41.31 (9.73)	40.21 (9.53)
tenure (month)	87.27 (86.98)	64.90 (74.35)	53.97 (59.07)	47.23 (55.98)	92.27 (96.14)	74.34 (86.21)	72.24 (78.65)	64.21 (71.70)
occupation change	0.17 (0.38)	0.15 (0.36)	0.15 (0.36)	0.17 (0.38)	0.10 (0.30)	0.11 (0.31)	0.09 (0.29)	0.11 (0.32)
occ. change (2-digit)	0.14 (0.35)	0.13 (0.34)	0.10 (0.30)	0.12 (0.32)	0.08 (0.28)	0.09 (0.29)	0.05 (0.22)	0.08 (0.28)
sector change	0.06 (0.23)	0.07 (0.26)	0.07 (0.26)	0.10 (0.30)	0.05 (0.22)	0.07 (0.25)	0.07 (0.25)	0.09 (0.28)
log occ. distance	2.21 (0.92)	2.35 (0.90)	2.40 (0.87)	2.49 (0.98)	2.22 (0.88)	2.30 (0.91)	2.28 (0.84)	2.51 (0.95)
employment share (in %)	12.8	31.8	17.8	37.6	16.7	32.3	20.8	30.1
observations	270k	670k	374k	791k	482k	931k	598k	878k

Note: the cells show the mean and standard deviation (in parenthesis) for selected employment transition indicators in 2003 (columns (1) to (4)) and in 2018 (columns (5) to (8)) for each of the discrete skill groups. Prices were adjusted by the official consumer price index (IPCA) to 2018 prices. Calculations apply weights for employment skill duration and hours worked based on the consecutive employment sample.

Likewise, the evolution of workers' age reflects, at least partly, the demographic transition in Brazil (IBGE 2018). All skill groups exhibited a higher average age in 2018 compared to 2003. Additionally, the most pronounced increase in average age occurred between 2013 and 2018, suggesting that firms primarily dismissed younger, less experienced workers during the economic recession. Workers in non-routine, cognitive-intensive occupations have longer employment tenure, increasing from 87.3 months in 2003 to 92.3 in 2018. RM workers are most prone to employment changes; that is to say that they present the lowest average tenure. Analogous to age, tenure shows a crescent trend that has accelerated since 2013 for all skill groups.

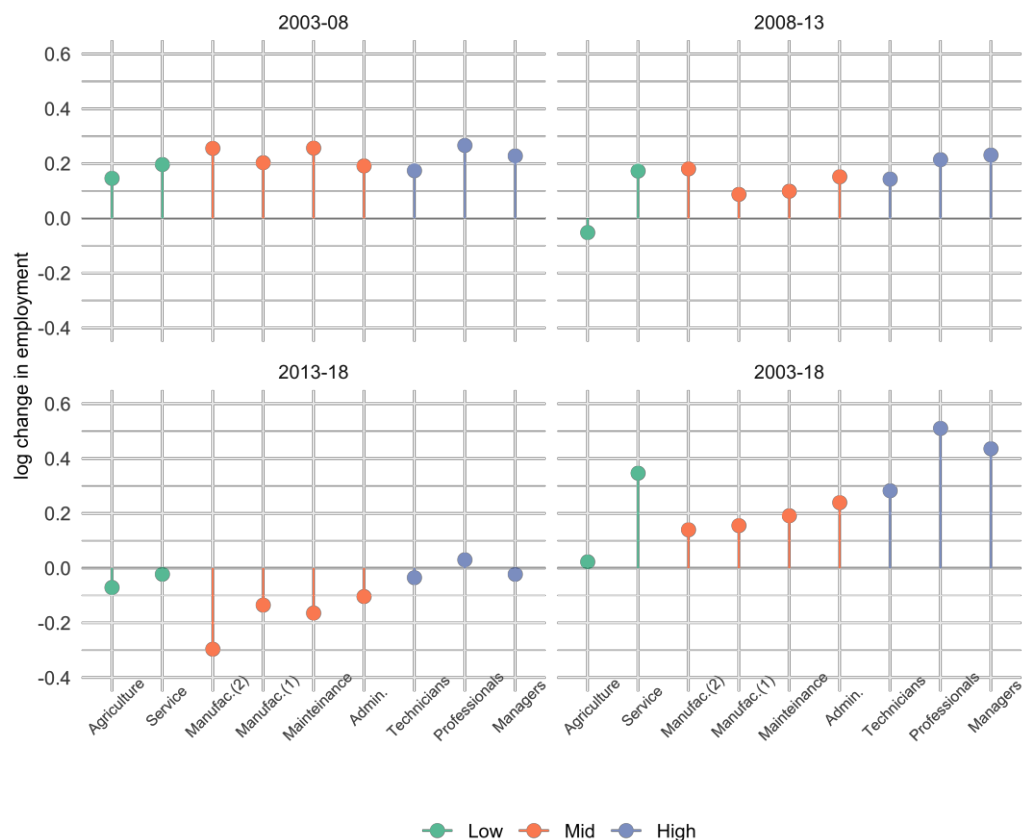
3.1.2 Employment changes and transitions

The Brazilian economy experienced a period of rapid GDP expansion between 2003 and 2009, with a mean growth rate of 4.2 per cent. The 2008 financial crisis did little harm to the economy, as GDP growth decelerated only slightly to a mean of 3.3 per cent between 2008 and 2013, before the economy plunged into a severe recession from 2014 onward (IBGE 2019b). The business cycle helps explain the following employment change patterns. Between 2013 and 2018, occupations in manufacturing suffered the strongest retraction of 30 per cent in employment, followed by maintenance workers and manufacturing of discrete processes. The recession was severe in sectors such as construction and transformation, falling to 85 per cent and 70 per cent of their production level in 2014, respectively (IFI 2018). Between 2003 and 2013, in contrast, all 1-digit occupation groups grew.

Figure 2 describes the changes in employment shares along with the wage distribution over the last two decades, following the seminal work of Autor, Katz and Kearney (2008). The total change and the breakdown for three sub-periods allow us to analyse whether Brazil experienced employment polarisation similar to developed countries. Only the most recent period, 2013 to 2018, shows employment growth in

occupations at the top and bottom of the wage distribution and a retraction in medium-income occupations. This is additional evidence of wage polarisation since the left and right tails of the wage distribution are increasing relative to its middle.

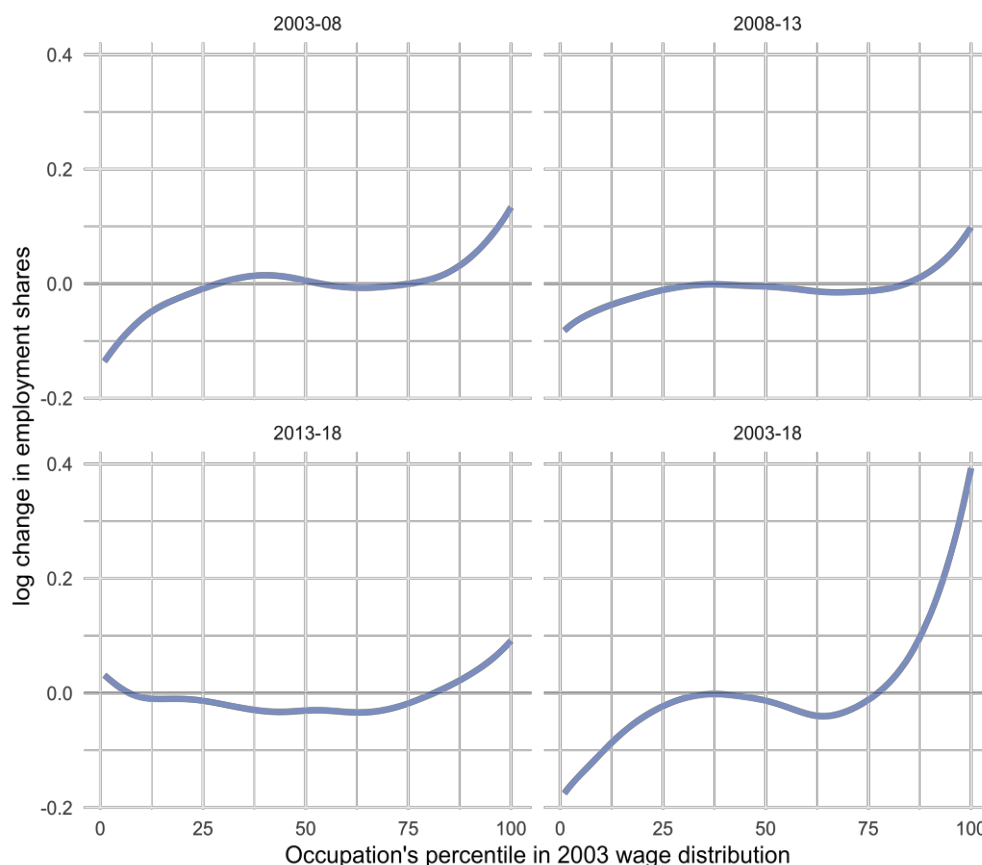
► **Figure 2. Employment changes by occupation group, 2003-2018.**



Note: the 1-digit occupations are ranked according to their mean hourly wage in 2003. See Appendix Table 1 for the precise definition of the occupation groups. Calculations are based on the consecutive employment sample and apply weights for employment duration and hours worked.

Over the last observation period (2013-2018), the Brazilian formal labour market resembles the United States labour market change between 1990 and 2000. In both countries, employment losses are most pronounced between the 40th and 60th wage percentile. The magnitude of the most extreme aggregate changes in a single percentile in Brazil range from +8 per cent to -3.5 per cent, while those reported by Autor, Katz and Kearney (2008) over a period twice as long are between -13 per cent and +16 per cent.

Figure 3. Employment changes over the wage distribution 2003-2018



Note: The figure shows the employment changes in 4-digit occupations ranked according to their 2003 mean wage using a locally weighted smoothing for the entire observation period, as well as three different subperiods, as indicated in the above four graphs. Wages are deflated to 2018 prices using the IPCA. Calculations are based on the entire RAIS sample and apply weights for employment skill duration and hours worked.

The polarization pattern is not seen when we consider other time intervals, like 2003 to 2008, 2008 to 2013, or the full period. Although high-skilled employment is continually expanding, low-skilled occupations experienced negative growth rates, and changes in the middle of the skill distribution were close to zero. Jaimovich and Siu (2020) and Hershbein and Kahn (2018) show that routine-biased technical change and jobless recoveries go hand in hand in the United States. Increasing automation of routine and manual tasks can accelerate these trends. That is, Brazilian firms may have used the recent recession as an “opportunity” to displace and substitute middle-skill workers.

3.2 Propensity to change occupation and sector

This section analyses how skill endowments influence the probability of workers moving into different occupations or sectors. That is, we aim to provide answers to the testable hypotheses H1, and H2.

Table 3 presents the results from the logit regressions where the binary dependent variables are the two definitions of occupation change and if a worker migrates to a different sector. The upper part of the table reports the marginal effects of the continuous skill variables. The lower part shows the predicted change probabilities for workers with average characteristics in each discrete skill group.¹²

¹²These regressions are based on the consecutive employment sample. This choice disregards occupation changes that occur after a period out of the labour market of more than one month. In this way, the estimations will neither be biased due to right censoring of employment spells nor confounded by the impacts that the worker experiences out of the labour market, such as unemployment stigma or educational upgrade. In even columns, the sample is additionally restricted to workers who change sectors.

The first column of Table 3 shows that higher NRC skills are positively related to the probability of moving to a different job. This relation is stronger than for any other type of skill, yet none of the marginal effects is statistically significant. Once the observable individual characteristics are taken into account, an average worker in a typical NRC job has an occupation change probability of 16 per cent. RC and NRM workers change occupations at slightly higher rates (19 per cent), whereas RM workers are most likely to switch occupations (22 per cent). Note that these predicted probabilities are higher and of a different order than the unconditional observed probabilities in Table 2. This means that characteristics such as workers' age, tenure, sectoral affiliation, employment contract type and firm size result in the lowest occupational and sectoral change probabilities for NRC workers. In contrast, NRM skills seem to be best employed in the same occupation, especially when one moves to a different sector.¹³

Table 3. Occupation change, sector change and skill endowments

dep. var.:	$\Delta Occ4$		$\Delta Occ2$		sector change
sample:	CE (1)	CE & $\Delta Sect.$ (2)	CE (3)	CE & $\Delta Sec.$ (4)	CE (5)
marginal effects					
NRC	0.541 (0.345)	0.340 (0.685)	0.304 (0.343)	0.494 (0.831)	0.119 (0.163)
NRM	0.448 (0.288)	-0.904 (0.511)	0.061 (0.300)	-1.307 (0.739)	0.482 (0.174)
RM	0.386 (0.277)	0.351 (0.561)	0.095 (0.248)	0.259 (0.602)	0.187 (0.120)
N	39,086,764	5,220,179	39,086,764	5,220,179	39,086,764
LL	-18,319,130	-3,092,822	-15,527,561	-3,384,799	-13,856,998
Pseudo R2	0.049	0.058	0.051	0.054	0.098
predicted probabilities					
NRC	0.160 (0.005)	0.623 (0.017)	0.128 (0.004)	0.530 (0.015)	0.089 (0.002)
RC	0.187 (0.009)	0.684 (0.020)	0.153 (0.005)	0.581 (0.016)	0.120 (0.003)
NRM	0.187 (0.006)	0.667 (0.015)	0.132 (0.006)	0.522 (0.013)	0.133 (0.004)
RM	0.219 (0.008)	0.688 (0.018)	0.158 (0.008)	0.568 (0.017)	0.164 (0.004)

Note: the upper part of the table presents the marginal effects of skill scores on the probability of either moving to a different occupation or sector applying logit regressions according to equation (4). The lower part of the table presents the predicted probabilities for workers from each of the four discrete skills with average characteristics. The dependent variables in columns (1) to (4) assume the value one if an occupation change between the current and the next employment spell is observed. Columns (1) and (2) use the 4-digit occupation classification, whereas columns (3) and (4) are based on the wider (2-digit) occupational classification. The estimation in columns (2) and (4) are similar to columns (1) and (3), respectively, but the sample is further restricted to workers who change sectors. In column (5), the dependent variable indicates whether the worker switches to a different sector and the consecutive employment (CE) spell sample is used as in columns (1) and (3). The estimations control for workers' tenure, gender, age and age squared, education, number of hours worked, type of employment contract, firm size and legal nature, as well as year, sector, and federal state fixed effects.

More precisely, column (2) shows that workers who use 10 per cent more NRM skills have a 9 per cent lower occupation change probability, *ceteris paribus*. By conditioning the sample based on sectoral change,

¹³ When we include fewer control variables in the estimations in Table 4 (only age, gender and year fixed effects), the coefficients tend to be lower but in the same order relative to each other. Thus, controlling for observable worker characteristics seems to disclose the differences among skills more clearly. Restricting the sample to male workers tends to magnify the coefficients' size. For example, apart from the significant coefficients in Table 4 the NRC coefficients in columns (1) and (3) become significant at the 10 per cent level.

we observe a jump in predicted probabilities to above 62 per cent for all skill groups. The differences in predicted probabilities across skill groups are less pronounced in this case. As in the previous estimations, NRC workers are most likely to maintain their previous occupation in the new sector, and RM workers are more likely to switch jobs as they change sectors. However, column (5) shows that NRM skills seem to boost sectoral switching the most. So, in this case, the work activity of NRM (predominantly service jobs) seems much more relevant than the sector in which it is applied. The high occupational and sectoral change probabilities of RM workers seem to stem from their unfavourable labour market position, including low wages, higher job instability and unemployment rates.

3.3 Consequences of occupation changes

Table 4 shows how skills relate to the distance between occupations and wages when workers switch occupations according to hypotheses H3 and H4. Recall that the regressions in Table 4: apply the continuous skill measures, and consider only job-to-job transitions.

The results in columns (1) to (2) in Table 4 indicate that having more non-routine skills decreases the occupational distance, and column (3) shows that workers with more RM skills move to more distant occupations once they change occupation and sector. Regarding wage changes following a job-to-job transition, columns (4) and (5) indicate that only the NRC skills show a positive relation, independent of whether the worker moves to an occupation with higher or lower *average* wage. Column (6) suggests that when workers change sectors and occupations, the differences between skill groups are insignificant.

It is important to note that the present approach does not identify marginal wage returns as Gonzaga and Guanziroli (2019) derive from Mincerian wage regressions. Evidence from Brazil and the United States indicates that NRC wages receive the highest marginal returns, particularly in agglomerations (Ehrl and Monasterio, 2019). The present approach only allows us to check whether otherwise observably equivalent workers instantly receive a systematically higher or lower wage in a novel or different occupation, depending on their current skill level. Another caveat is that the present estimates are conditional on changing jobs, inside or in another company. RAIS captures changes in job titles and wages annually, even if the worker remains with the same employer.

Moving forward to hypothesis H5, we explore how skills are related to the formal labour market exit probability, given that the worker re-enters at a later time. The reader should bear in mind that the period out of the labour market may either be unemployment, voluntary time-out or in the informal sector. Therefore, temporary illness or injury leave is not considered a labour market exit.

► **Table 4. Distance after occupation changes**

dep. var.:	log occupational distance			log wage change			hazard rate
sample:	$\Delta\text{Occ4.}\uparrow$	$\Delta\text{Occ4.}\downarrow$	$\Delta\text{Sect.}$	$\Delta\text{Occ4.}\uparrow$	$\Delta\text{Occ4.}\downarrow$	$\Delta\text{Sect.}$	full sample
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
NRC	-4.345 (2.950)	-4.812 (2.893)	0.021 (2.393)	0.587 (0.177)	0.385 (0.214)	0.296 (0.314)	0.730 (1.085)
NRM	-2.933 (3.261)	-6.489 (2.519)	-1.989 (2.391)	0.183 (0.149)	-0.226 (0.202)	-0.030 (0.310)	2.099 (1.145)
RM	2.241 (2.255)	-0.761 (2.397)	3.842 (1.811)	0.161 (0.150)	0.302 (0.198)	0.285 (0.255)	1.930 (0.834)
N	4,231,257	3,371,208	3,536,374	4,231,257	3,371,208	3,536,374	50,082,074
F	38	19	26	341	186	320	
R2	0.07	0.059	0.07	0.032	0.040	0.061	
Informat. criteria							LL -7,247,139 AIC 14,494,504 BIC 14,496,282

Note: The table presents the estimated coefficient of skill measures (non-routine cognitive (NRC), routine cognitive (RC), non-routine manual (NRM), and routine manual (RM)). The second line indicates whether the sample consists of workers changing: to an occupation with higher average wage ($\Delta\text{Occ4.}\uparrow$; columns (1) and (4)); to an occupation with lower average wage ($\Delta\text{Occ4.}\downarrow$; columns (2) to (5)); to a different sector ($\Delta\text{Sect.}$; columns (3) and (6)). The estimations control for workers' tenure, gender, age and age squared, education, number of hours worked, type of employment contract, firm size, legal nature, as well as year, sector, and federal state fixed effects. Standard errors in parenthesis are clustered at the occupation level.

Column 7 on Table 4 displays the estimated coefficients ξ according to the model specified in equation (6). That is, variables with positive coefficients increase a worker's hazard rate, *ceteris paribus*. It indicates that workers who use manual skills intensively have a greater exit risk than the reference category of RC workers. For a different interpretation of the results, recall that the hazard rate representation can be derived as $\exp(\xi)$. So, for example, the elasticity of the hazard rate with respect to RM skill is equal to $6.9 = \exp(1.93)$. Occupations that require ten p.p. more RM skills have a 69 per cent higher hazard rate.

4 Final remarks

This paper provided evidence that skill endowments are related to the probability of a worker leaving the formal labour market and transitioning to a different occupation or sector in contemporary Brazil. These estimates were possible because we introduced two methods of analysing occupations. The first one calculated the distance between pairs of occupations based on the description of the tasks they entail, while the second method quantified the skills required for each occupation. These methods overcame limitations of previous approaches and can be replicated in other countries that provide task descriptions for occupations.

Workers' skills also seem to play a fundamental role in explaining the distance in terms of tasks between two occupations and related wage changes. By and large, we document that NRC workers are in the most privileged position in all the analysed areas. In particular, their skills protect them relatively well against job losses, even during an economic recession. The evidence supports the RBTC hypothesis and is in line with the fundamental changes brought by computerization and automation technologies. That is, at least in recent years, we see a falling employment share of middle-skill workers, intensive in routine activities, in favour of low-skill jobs not worth automating and high-skill workers complemented by automation.

Our assessment of the Brazilian labour market covers two decades of profound changes and economic crisis, but it is mostly descriptive. Future research can improve upon differentiating the diverse mechanisms

by which different shocks – related to technology, outsourcing or production – impact the labour market in developing countries. It also remains to be seen whether the documented labour market polarisation is permanent or temporary.

Nevertheless, some policy lessons can be drawn from our findings. In line with previous papers, the data indicate that (routine) manual workers are heavily impacted by economic downturns, even in the long run. To minimize such impacts, efforts to stimulate human capital accumulation for this population group are highly recommended. Studies based on the task approach found that training programmes have the potential to increase workers' performance in non-routine analytical and soft skills (Tamm 2018; Görlitz and Tamm 2016). However, the impact of those programmes on employment or wages does not seem as promising. On the one hand, international evidence shows that success is not trivial and depends on the details of the design and implementation of these programmes. On the other hand, training measures tend to be more successful in low- and middle-income countries (Kluve et al., 2017), particularly when firms are involved in defining the content of the courses (Petterini, 2016; O'Connell et al., 2017).

Studies for automation in the Brazilian labour market consider that at least 45 per cent of employment, most of it intensive in low-skill manual occupations, is susceptible to automation in the near future (Albuquerque et al., 2019; Adamczyk, 2021). However, at least until now, automation technologies in Brazil need to be seen as prospects, not showing clear trends of employment destruction and massive technological unemployment, as foreseen by Autor (2015) or Willcocks (2020).

References

- Acemoglu, Daron, and David Autor. 2011. "Skills, Tasks and Technologies: Implications for Employment and Earnings". In *Handbook of Labor Economics*, 4:1043–1171.
- Acemoglu, Daron, and Pascual Restrepo. 2017. "Robots and Jobs: Evidence from US Labor Markets". National Bureau of Economic Research Working Paper No. 23285. <https://doi.org/10.3386/w23285>
- Adamczyk, Willian Boschetti. 2021. "Ensaio Sobre as Tecnologias de Automação No Mercado de Trabalho Brasileiro". PhD thesis, Escola de Negócios - Pontifícia Universidade Católica do Rio Grande do Sul.
- Adamczyk, Willian Boschetti, Leonardo Monasterio, and Adelar Fochezatto. 2021. "Automation in the Future of Public Sector Employment: The Case of Brazilian Federal Government". *Technology in Society* 67: 101722.
- Aggarwal, Charu C. 2015. *Data Mining: The Textbook*. New York: Springer.
- Aggarwal, Charu C, Alexander Hinneburg, and Daniel A Keim. 2001. "On the Surprising Behavior of Distance Metrics in High Dimensional Space". In *International Conference on Database Theory*, 420–34. New York: Springer.
- Albuquerque, Pedro Henrique Melo, Cayan Atreio Portela Bárcena Saavedra, Rafael Lima de Moraes, and Yaohao Peng. 2019. "The Robot from Ipanema Goes Working: Estimating the Probability of Jobs Automation in Brazil". *Latin American Business Review* 20 (3): 227–48.
- Almeida, Rita, Carlos Henrique Corseuil, and Jennifer Poole. 2017. "The Impact of Digital Technologies on Routine Tasks: Do Labor Policies Matter?" SSRN Scholarly Paper 6560. World Bank Policy Research Working Paper.
- Almeida, Rubiane Daniele Cardoso, Philipp Ehrl, and Tito Belchior Silva Moreira. 2021. "Social and Economic Convergence Across Brazilian States Between 1990 and 2010." *Social Indicators Research* 157: 225–46.
- Arellano-Bover, Jaime. 2020. "The Effect of Labor Market Conditions at Entry on Workers' Long-Term Skills". *Review of Economics and Statistics*, 1–45.

- Ariza, John, and Josep Lluís Raymond Bara. 2020. "Technological Change and Employment in Brazil, Colombia and Mexico: Which Workers Are Most Affected?" *International Labour Review* 159 (2): 137–59.
- Arntz, Melanie, Terry Gregory, and Ulrich Zierahn. 2017. "Revisiting the risk of automation". *Economics Letters* 159: 157–160.
- Autor, David H. 2015. "Why Are There Still So Many Jobs? The History and Future of Workplace Automation". *Journal of Economic Perspectives* 29 (3): 3–30.
- Autor, David H, Lawrence F Katz, and Melissa S Kearney. 2008. "Trends in US Wage Inequality: Revising the Revisionists". *Review of Economics and Statistics* 90 (2): 300–323.
- Autor, David H, Frank Levy, and Richard J Murnane. 2003. "The Skill Content of Recent Technological Change: An Empirical Exploration". *The Quarterly Journal of Economics* 118 (4): 1279–1333.
- Bachmann, Ronald, Merve Cim, and Colin Green. 2019. "Long-Run Patterns of Labour Market Polarization: Evidence from German Micro Data". *British Journal of Industrial Relations* 57 (2): 350–76.
- Carrillo-Tudela, Carlos, Bart Hobijn, Powen She, and Ludo Visschers. 2016. "The Extent and Cyclicity of Career Changes: Evidence for the UK". *European Economic Review* 84 (May): 18–41.
- Comissão Nacional de Classificação - CONCLA. 2019. "Classificação Brasileira de Ocupações – CBO". Instituto Brasileiro de Geografia e Estatística, Available at: <https://concla.ibge.gov.br/classificacoes/por-tema/ocupacao/classificacao-brasileira-de-ocupacoes.html>.
- Cortes, Guido Matias. 2016. "Where Have the Middle-Wage Workers Gone? A Study of Polarization Using Panel Data". *Journal of Labor Economics* 34 (1): 63–105.
- Cortes, Guido Matias, Nir Jaimovich, Christopher J Nekarda, and Henry E Siu. 2020. "The Dynamics of Disappearing Routine Jobs: A Flows Approach". *Labour Economics* 65: 101823.
- Detoni, Otávio Florentino, Ricardo Freguglia, and Carlos Henrique Corseuil. 2020. "Prêmio Salarial Associado Às Competências Dos Trabalhadores No Brasil: Uma Análise Com Dados Em Painel (2003–2013)". In 48 Encontro Nacional de Economia. ANPEC.
- Dix-Carneiro, Rafael, Pinelopi K Goldberg, Costas Meghir, and Gabriel Ulyssea. 2021. "Trade and Informality in the Presence of Labor Market Frictions and Regulations". Working Paper 28391. Working Paper Series. National Bureau of Economic Research.
- Dix-Carneiro, Rafael, and Brian K. Kovak. 2019. "Margins of labor market adjustment to trade". *Journal of International Economics* 117 (C): 125–42.
- Ehrl, Philipp. 2018. "Task Trade and Employment Patterns: The Offshoring and Onshoring of Brazilian Firms". *Journal of International Trade & Economic Development* 27 (3): 235–66.
- Ehrl, Philipp, and Leonardo Monasterio. 2021. "Spatial Skill Concentration Agglomeration Economies". *Journal of Regional Science* 61 (1): 140–61.
- Ehrl, Philipp. 2019. "Skill Concentration and Persistence in Brazil". *Regional Studies* 53: 1544–54.
- Firpo, Sergio, and Alysson Portella. 2019. Decline In Wage Inequality In Brazil: A Survey. World Bank, Washington, DC. <https://doi.org/10.1596/1813-9450-9096>.
- Frey, Carl Benedikt, and Michael A Osborne. 2017. "The Future of Employment: How Susceptible Are Jobs to Computerization?" *Technological Forecasting and Social Change* 114: 254–80.
- Gathmann, Christina, and Uta Schönberg. 2010. "How General Is Human Capital? A Task-Based Approach". *Journal of Labor Economics* 28 (1): 1–49.
- Gonzaga, Gustavo, and Tomás Guanziroli. 2019. "Returns to Experience Across Tasks: Evidence from Brazil". *Applied Economics Letters* 26 (20): 1718–23.
- Goos, Maarten, and Alan Manning. 2007. "Lousy and Lovely Jobs: The Rising Polarisation of Work in Britain". *The Review of Economics and Statistics* 89 (1): 118–33.

- Goos, Maarten, Alan Manning, and Anna Salomons. 2014. "Explaining Job Polarisation: Routine-Biased Technological Change and Offshoring". *American Economic Review* 104 (8): 2509–26.
- Görlitz, Katja, and Marcus Tamm. 2016. "The Returns to Voucher-Financed Training on Wages, Employment and Job Tasks". *Economics of Education Review* 52: 51–62.
- Graetz, Georg, and Guy Michaels. 2017. "Is Modern Technology Responsible for Jobless Recoveries?" *American Economic Review* 107 (5): 168–73.
- Herdeiro, Renato., Naércio Menezes-Filho, and Bruno Komatsu. 2019. "Explicando a Evolução Dos Salários Relativos Por Grupos de Qualificação No Brasil". In 47 Encontro Nacional de Economia. ANPEC.
- Hershbein, Brad, and Lisa B. Kahn. 2018. "Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings". *American Economic Review* 108 (7): 1737–72.
- IBGE, Instituto Brasileiro de Geografia e Estatística. 2018. "Projeções Da População Do Brasil e Unidades Da Federação Por Sexo e Idade – 2010-2060".
- IBGE, Instituto Brasileiro de Geografia e Estatística. 2019a. "Censo Agropecuário de 2017".
- IBGE, Instituto Brasileiro de Geografia e Estatística. 2019b. "Sistema de Contas Nacionais. Tabela 6 – Produto Interno Bruto, Produto Interno Bruto Per Capita, População Residente e Deflator – 1996-2018".
- IFI, Instituição Fiscal Independente. 2018. "Relatório de Acompanhamento Fiscal, Nº 35, Dezembro de 2019".
- Jaimovich, Nir, and Henry E. Siu. 2020. "Job Polarisation and Jobless Recoveries". *The Review of Economics and Statistics* 102 (1): 129–47.
- Kluve, Jochen, Susana Puerto, David Robalino, Jose Manuel Romero, Friederike Rother, Jonathan Stoeterau, Felix Weidenkaff, and Marc Witte. 2017. "Interventions to Improve the Labour Market Outcomes of Youth: A Systematic Review of Training, Entrepreneurship Promotion, Employment Services and Subsidized Employment Interventions". *Campbell Systematic Reviews* 13 (1): 1–288.
- Lancaster, Tony. 1979. "Econometric Methods for the Duration of Unemployment". *Econometrica*, 939–56.
- Maciente, Aguinaldo Nogueira. 2013. "The Determinants of Agglomeration in Brazil: Input-Output, Labor and Knowledge Externalities". Ph.D. Thesis, Department of Agricultural; Consumer Economics; University of Illinois at Urbana-Champaign.
- Maloney, William F., and Carlos Molina. 2019. *Is Automation Labor-Displacing in the Developing Countries, Too? Robots, Polarisation, and Jobs*. World Bank.
- O'Connell, Stephen D, Lucas Ferreira Mation, Joao Bevilaqua Teixeira Basto, and Mark A Dutz. 2017. "Can Business Input Improve the Effectiveness of Worker Training? Evidence from Brazil's Pronatec-MDIC". 8155. Evidence from Brazil's Pronatec-MDIC (July 31, 2017). World Bank Policy Research Working Paper.
- Petterini, Francis Carlo. 2016. "Uma Avaliação Econômica Do Plano Setorial de Qualificação (PLANSEQ). *Economia Aplicada* 20 (3): 173–94.
- Puggioni, D., Calderón, M., Zurita, A. C., Bujanda, L. F., González, J. A. I., & Jaume, D. Forthcoming. "Inequality, Income Dynamics, and Worker Transitions: The Case of Mexico. *Quantitative Economics*".
- Qaiser, Shahzad, and Ramsha Ali. 2018. "Text Mining: Use of TF-IDF to Examine the Relevance of Words to Documents". *International Journal of Computer Applications* 181 (1): 25–29.
- Samaniego de la Parra, Brenda, and Viegelahn, Christian. 2021. "Estimating Labour Market Transitions from Labour Force Surveys the Case of Viet Nam", ILO Working Paper 35.
- Simpson, Edward H. 1949. "Measurement of Diversity". *Nature* 163 (4148): 688–88.

- Spitz-Oener, Alexandra. 2006. "Technical Change, Job Tasks, and Rising Educational Demands: Looking Outside the Wage Structure". *Journal of Labor Economics* 24 (2): 235–70.
- Tamm, Marcus. 2018. "Training and Changes in Job Tasks". *Economics of Education Review* 67 (December): 137–47.
- Ulyssea, Gabriel. 2020. "Informality: Causes and Consequences for Development". *Annual Review of Economics* 12 (1): 525–46.
- Wiczer, David G. 2015. "Long-Term Unemployment: Attached and Mismatched?" Federal Reserve Bank of St. Louis. Working Paper 2015-042A.
- Willcocks, Leslie. 2020. "Robo-Apocalypse Cancelled? Reframing the Automation and Future of Work Debate". *Journal of Information Technology* 35 (4): 286–302.